Chapter 7 • Coordinate Geometry

# Solved Exercise 7.1

**1. Describe the location in the plane of P(x,y) for each condition.**

(i) $x>0$: the open right half-plane, excluding the y-axis.
**right half-plane**
(ii) $x>0$ and $y>0$: Quadrant I.
**QI**
(iii) $x=0$: the y-axis.
(iv) $y=0$: the x-axis.
(v) $x>0$ and $y<0$: Quadrant IV only; both strict inequalities exclude the axes.
**QIV only**
(vi) $y=0$ and $x=0$: the origin (0,0).
**origin**
(vii) $x=y$: the line $y=x$, bisecting Quadrants I and III.
**y=x**
(viii) $x\geq 3$: the vertical line $x=3$ together with the half-plane to its right.
**x≥3 region**
(ix) $y>0$: the open half-plane above the x-axis.
**above x-axis**
(x) x and y have opposite signs: Quadrants II and IV, excluding the axes.
**QII and QIV.**

**2. Find the distance between each pair of points using $d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$.**

(i) A(6,7),B(0,−2): $d = \sqrt{(0-6)^2+(-2-7)^2}$
$= \sqrt{(36+81)}$
$= \sqrt{117}$
$= 3\sqrt{13}$.
(ii) C(−5,−2),D(3,2): $d = \sqrt{(3+5)^2+(2+2)^2}$
$= \sqrt{(64+16)}$
$= \sqrt{80}$
$= 4\sqrt{5}$.
(iii) L(0,3),M(−2,−4): $d = \sqrt{(-2-0)^2+(-4-3)^2}$
$= \sqrt{(4+49)}$
$= \sqrt{53}$.
(iv) P(−8,−7),O(0,0): $d = \sqrt{(0+8)^2+(0+7)^2}$
$= \sqrt{(64+49)}$
$= \sqrt{113}$.

**3. For each pair, find distance and midpoint.**

(a) A(3,1), B(−2,−4): $d = \sqrt{(5^2+5^2)}$
$= 5\sqrt{2}$
M=$(\frac{1}{2},\frac{-3}{2})$.
(b) A(−8,3), B(2,−1): $d = \sqrt{(10^2+4^2)}$
$= 2\sqrt{29}$
M=(−3,1).
(c) A(−√5,−1/3), B(−3√5,5): $d = \sqrt{[20+(16/3)^2]}$
$= \frac{2\sqrt{109}}{3}$
M=$(-2\sqrt{5},\frac{7}{3})$.

**4. Which points are 15 units from the origin?**

(i) $\sqrt{(176+7^2)} = \sqrt{225}$
$= 15$: yes.
(ii) $\sqrt{(10^2+10^2)}=10\sqrt{2}\neq 15$: no
(iii) $\sqrt{(1^2+15^2)}=\sqrt{226}\neq 15$: no.

**5. Prove the three triangle/quadrilateral statements.**

(i) $AB^2=4$, $BC^2=12$, $AC^2=16$. Since $AB^2+BC^2=AC^2$, the triangle is right-angled at B.
(ii) $AB^2=41$ and $BC^2=41$, while $AC^2=2$. Hence AB=BC: isosceles.
(iii) AB=DC=(−7,1) and AD=BC=(−1,−7). Both pairs of opposite sides are equal and parallel: parallelogram.

**6. Find h for a right angle at A, with A(√3,−1), B(0,2), C(h,−2).**

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AB=(−√3,3), AC=(h−√3,−1). Perpendicularity gives AB·AC=0.
−√3(h−√3)−3 = −√3h
= 0.

**7. Find h so A(−1,h), B(3,2), C(7,3) are collinear.**

Slope BC=$\frac{3-2}{7-3}=\frac{1}{4}$ Slope AB=$\frac{2-h}{4}$ must also equal $\frac{1}{4}$

**h=1.**

**8. A(−5,−2) and B(5,−4) are ends of a diameter. Find centre and radius.**

Centre = midpoint
= (0,−3). Diameter
= √(10²+2²)
= 2√26.

**Centre (0,−3); radius √26 units.**

**9. Find h for a right angle at A, with A(h,1), B(2,7), C(−6,−7).**

(2−h)(−6−h)+6(−8)=0
⇒ h²+4h−60=0.
(h+10)(h−6)=0.

**h=−10 or h=6.**

**10. Find side midpoints of A(9,3), B(−7,7), C(−3,−7), D(5,−5), and prove the midpoint quadrilateral is a parallelogram.**

Midpoints: E(1,5), F(−5,0), G(1,−6), H(7,−1).
EF=(−6,−5)=−GH and FG=(6,−6)=−HE. Opposite sides are parallel and equal.

**E(1,5), F(−5,0), G(1,−6), H(7,−1); EFGH is a parallelogram.**

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